

Dasymetric Disaggregation of Employment and Housing Data

A set of linked ArcGIS Python Toolboxes that break coarse county-level employment and housing statistics down to fine-grained spatial zones using land-use weighted dasymetric mapping

The Problem

Government statistics like employment counts and household totals are usually published at coarse geographies, often no finer than a county, but spatial analysis needs those totals broken down to much finer sub-county zones. Simply dividing a county total evenly across its zones assumes uniform density, which is wrong, since dense urban zones and rural zones don't hold the same amount of activity per acre. Getting a defensible fine-grained estimate runs into several real problems.

- Employment data arrives as establishment-level records with wildly varying location precision. Some establishments carry rooftop-level geocoding, others only resolve to a ZIP code or not at all.
- A meaningful share of establishment records are administrative noise rather than real workplace locations: multi-location headquarters that would double-count employment if kept as-is, remote or telecommute listings with no fixed workplace, and temp staffing agencies whose payroll doesn't reflect where people actually work day to day.
- Household counts come from a separate Census survey product, delivered as a raw CSV published at the county level only, with margin-of-error columns that need to be checked rather than estimates taken at face value.
- Whatever fine-grained values get produced have to add up exactly to the original county totals. An estimate that doesn't reconcile back to the published control number isn't defensible.
- Land use data quality isn't uniform across a large study area, so a disaggregation weight based on observed development works well in some areas but needs a different, less data-intensive fallback elsewhere.

My Approach

I built three linked **ArcGIS Python Toolboxes** forming a disaggregation pipeline, employment preparation, household allocation, and a final composite density calculation, so each stage can be run and reviewed on its own before feeding the next.

Hybrid point-in-polygon plus dasymetric fallback for employment. Employment establishment records are cleaned first, removing headquarters duplicates and telecommuter listings, and redistributing temp staffing employment based on documented share splits. Each record is then classified into standard categories via NAICS code lookups

against a configurable classification table, with explicit overrides for government and K-12 employers. Records with reliable point-level geocoding are assigned directly to their containing zone through a spatial join. Records with only coarse geocoding fall back to proportional dasymetric allocation, spread across a county's zones weighted by each zone's share of developed land in the matching land use category.

Dasymetric weighting for households from a coarser Census product. Household totals come from an American Community Survey table published at the county level. I built a parser for the raw Census CSV export, handling its specific label rows, geography ID format, and margin-of-error columns, then allocated each county's total down to its zones proportionally by residential developed acreage, the same dasymetric logic used as the employment fallback, but as the primary method here since no point-level household data exists.

Two-tier weighting for reliability. Not every area has land use data of the same quality. For most zones, the allocation weight is real, observed developed acreage. For zones flagged as lower confidence, I used geographic area as a fallback weight instead of assuming the observed acreage is representative, with a majority-rule check deciding which weighting a zone's county uses.

Exact reconciliation to control totals. Proportional allocation produces fractional values that don't sum to whole numbers on their own. I round every zone down, then distribute the rounding remainder to the zones with the largest fractional leftover, one at a time, until each county's zone values sum to exactly its original control total. The estimate is defensible in aggregate by construction, not just approximately close.

Composite activity density as the final output. Once employment and household estimates exist at the zone level, I combine them with each zone's acreage into a single composite activity density metric, then bucket every zone into ten classes so each class represents an equal share of total activity rather than an equal number of zones, which better reflects how unevenly real activity concentrates across a landscape.

Validation before anything is trusted. Every allocation run is checked against its own control totals (do the zone values still sum exactly to the county number), sanity bounds (is the density per acre in a plausible range), and a rule that zero-development zones never receive a nonzero allocation. Every run also produces a full log covering data cleaning decisions, classification counts, and validation results.

Technical Highlights

- **Language/Platform:** Python, ArcGIS Pro ([arcpy](#)) Python Toolboxes, pandas and numpy for tabular processing
- **Data sources:** State employment establishment micro-data, American Community Survey housing tables, land use acreage from a prior parcel classification stage

- **Data cleaning:** Duplicate and headquarters record removal, telecommuter filtering, proportional redistribution of ambiguous employment categories
- **Classification:** Configuration table driven category mapping with explicit override rules for known special cases
- **Spatial methods:** Point-in-polygon spatial join for precisely geocoded records, dasymetric proportional allocation for everything else, majority-rule tiered weighting for reliability
- **Reconciliation:** Exact integer rounding reconciliation so disaggregated values always sum to published control totals
- **Output:** A composite activity density metric with data-driven class breakpoints, plus full audit logs and multi-check validation for every run

Why It Matters

Fine-grained employment and household estimates only earn trust if they can be traced back to a real methodology and reconcile to numbers a data consumer can independently verify. Building the disaggregation to fall back gracefully when precise data isn't available, reconcile exactly to published control totals, and log every classification and validation decision means the fine-grained output can be handed to someone else and defended, not just used as an internal black box number.